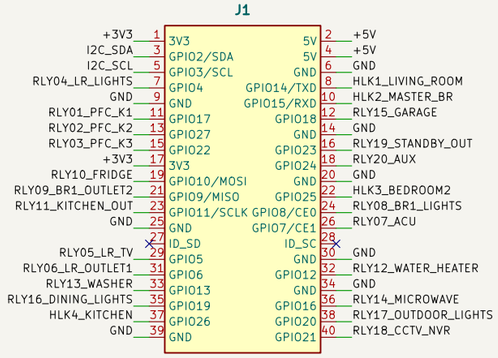
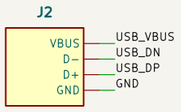
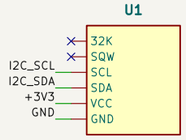
RASPBERRY PI 5 DOMAIN (relays, presence sensors, RTC)



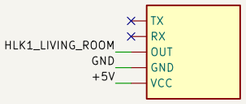
Raspberry Pi 5 — 40-pin GPIO header



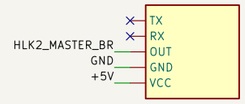
Raspberry Pi 5 — USB-A port



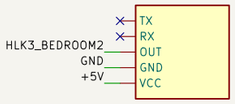
DS3231 RTC module (I2C 0x68)



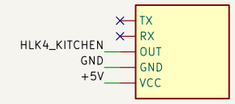
HLK-LD2410B zone1 — Living Room



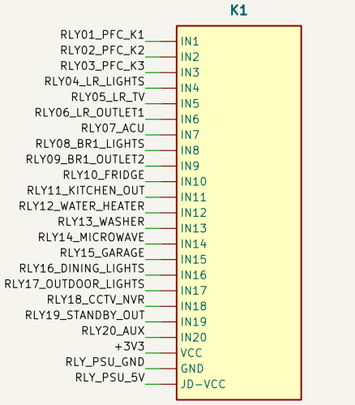
HLK-LD2410B zone2 — Master BR



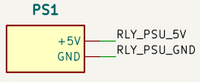
HLK-LD2410B zone3 — Bedroom2



HLK-LD2410B zone4 — Kitchen



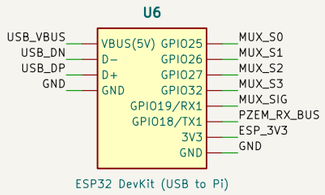
20-ch opto relay board (active-LOW)



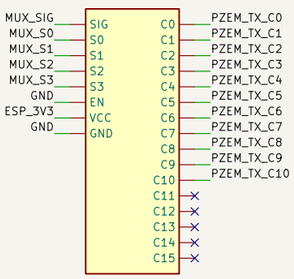
5V >=2A relay coil supply

WIRING NOTES
1. Same label name = same wire. Connect every pin that shares a label.
   Full pin-to-pin list: NETLIST.md
2. J1 pin numbers are PHYSICAL header pins (not BCM).
3. K1 relay board: REMOVE the VCC/JD-VCC jumper.
   VCC = PI 3V3 (opto LEDs); JD-VCC + GND = PS1 (5V >=2A).
   Do NOT tie PI GND to K1 GND (keeps opto isolation).
4. Relays are active-LOW: GPIO LOW = relay ON.
   RLY01-03 drive the PFC capacitor-bank contactors.
5. GPIO14/15 (HLK zones 1-2) are UART pins:
   disable serial console + serial hardware.
6. Keep SPI (GPIO7-11) and audio/PWM overlays
   (GPIO12/13/18/19) disabled.
7. Boot check: GPIO9-27 pull down at power-up, so relays
   may click ON until app.py starts. Fix if needed:
   config.txt: gpio=9-13,16-24,27=op,dh
8. ESP32 side is all 3.3V: PZEM TTL '5V' pins and mux
   VCC come from ESP32 3V3 (ESP_3V3). No level shifter.
9. PZEM_RX_BUS: ESP32 TX1 -> RX of ALL PZEMs. The mux
   returns only the selected PZEM TX to ESP32 RX1.
10. Mux EN = GND (always on). C11-C15 unused.
11. Control wiring only: mains side not shown.

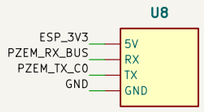
ESP32 DOMAIN (11x PZEM-004T via CD74HC4067, all 3.3V logic)



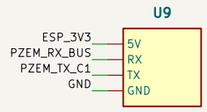
ESP32 DevKit (USB to Pi)



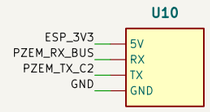
CD74HC4067 mux breakout



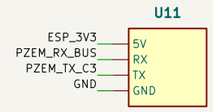
PZEM #0 Main Panel



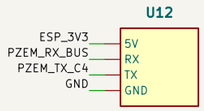
PZEM #1 Branch 1 (R4)



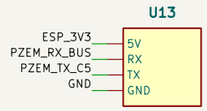
PZEM #2 Branch 2 (R5)



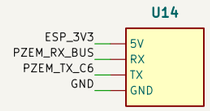
PZEM #3 Branch 3 (R6)



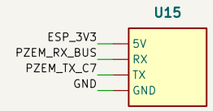
PZEM #4 Branch 4 (R7)



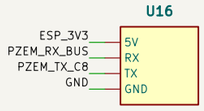
PZEM #5 Branch 5 (R8)



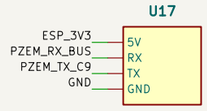
PZEM #6 Branch 6 (R9)



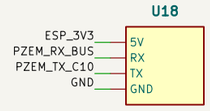
PZEM #7 Branch 7 (R10)



PZEM #8 Branch 8 (R11)



PZEM #9 Branch 9 (R12)



PZEM #10 Branch 10 (R13)

Control (low-voltage) wiring only
Generated by generate_schematic.py from app.py + ESP32 .ino
REMS / RECMS Thesis

Sheet: /
File: rems-control-wiring.kicad_sch

Title: REMS — Control Wiring Schematic

Size: A3	Date: 2026-10-14	**Rev: 1.0**
KiCad E.D.A. 10.0.6		Id: 1/1